Title: Design and Implementation of a Blockchain-based Electronic Voting System

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ABSTRACT

With the advancement of society and its people's democratic awareness, voting as a vital channel of democracy must meet contemporary society's high expectations; new technology and tactics must be employed for a better voting experience, and to replace the inefficient and time-consuming traditional paper voting. Blockchain technology is now impacting every sector where data security and integrity is important. In this paper, a blockchain based electronic voting system is designed and implemented to provide a secure and open election environment in which users may vote only once and the result of the election is shown without being changed after the election. Working blockchains will prevent tampering with votes and third-party manipulation of the processes. Blockchain will eliminate a lot of issues related to online voting. To avoid different forms of voting fraud, casted votes may be documented and reported using blockchain technology and voters can utilize this method to cast their ballots without revealing their identity or political views.

CHAPTER ONE

- INTRODUCTION

1.1 BACKGROUND OF THE STUDY

Democracy is generally regarded as the most effective form of government. Democracy as a system protects people from absolute power abuse and guarantees citizens the rights and freedoms of the majority. There are some critical problems in the field of democratic elections, such as vote counting and verification, which are handled by involving various figures and devices in the conventional phase of a voting process. The vote represents a fundamental element of any collaboration between people, and it can be done in a variety of ways, including online voting, raising hands to ballot boxes, and so on. Electronic voting as a promising voting system has increasingly gained widespread interest as Internet and cryptography technology advances. Since its first use as punched-card ballots in the 1960s, e-voting systems have achieved remarkable progress with their adaption using internet technologies (Gobel *et al.*, 2015). However, e-voting systems must adhere to specific benchmark parameters to facilitate their widespread adoption. These parameters include anonymity of the voter, integrity of the vote, and non-repudiation among others.

Electronic voting is defined as using electronic means to help implement voting and counting. Electronic voting technology, as compared to conventional paper-based voting, can speed up the counting of ballots, minimize personnel costs and to some degree, ensure electoral fairness and protect voters' interests. Thus, a valuable and successful electronic voting system must be secure, transparent, anonymous, and efficient. However, most e-voting platforms build their voting platforms based on a centralized server-client model. These platforms adopt central agencies to record and calculate votes. The crucial challenge for centralized voting systems is that voters cannot verify their ballots since their votes may be tampered with by administers. Another severe obstacle is the risk of central servers' downtime as centralized voting systems are weak in terms of security, anonymity, and process transparency. At this stage, the idea of using blockchain technology to distribute an electoral register open to all citizens is becoming increasingly popular.

The term blockchain was first described back in 1991 by a group of researchers who wanted to create a tool to timestamp digital documents so that they could not be backdated or changed. Further, the technique was adopted and reinvented by Satoshi Nakamoto. In 2009, Nakamoto created the first cryptocurrency, the blockchain-based project called Bitcoin. Blockchain is a distributed database solution that maintains a continuously growing list of data records that are confirmed by the nodes participating in it. The data is recorded in a public ledger, including information on every transaction ever completed. Blockchain is a decentralized solution that does not require any third-party organization in the middle (Yli-Huumo *et al*., 2016). Each block is connected to a previous block by a hash, thus enabling traceability; meanwhile, cryptography is used to guarantee that the block data cannot be tampered with and cannot be forged. If any changes are made in the block, the cryptographic hash would change immediately indicating the change in the data which may be due to malicious activity. Therefore, due to its strong foundations in cryptography, blockchain has been increasingly used to mitigate unauthorized transactions across various domains (Narayanan *et al.*, 2015).

Blockchain technology can play a key role in the domain of electronic voting due to the inherent nature of preserving anonymity, maintaining a decentralized and publicly distributed ledger of transactions across all the nodes, while votes are usually recorded, organized, counted, and verified by a central authority, a blockchain-based electronic voting system could decentralize controls, making voters take over certain tasks while retaining a copy of the electoral register (Fusco, 2018). This would lead to the elimination of illegitimate votes because each voter has a copy of the voting history that cannot be altered and whose changes can be easily verified. An illegitimate vote already registered or associated with an invalid register, cannot therefore be added because the other voters would be able to see that it is not compatible with the established rules. This makes blockchain technology very efficient to deal with the threat of utilizing a voting token more than once and the attempt to influence the transparency of the result.

1.2 STATEMENT OF THE PROBLEM

Most elections require voters to physically report to a polling booth and cast their ballot with a paper vote. While the procedure was more or less innocuous in developed economies until COVID-19 hit, booth rigging is far too widespread in poorer and third-world countries. Following the anti-trust backlash from the general public, a substantial number of nations have moved to electronic voting machines.

There are several advantages to remote (electronic) voting. It is assumed that they are more convenient for end-users because people can vote without leaving home; this increases the activity of voters. Maintenance of electronic voting is cheaper: instead of permanently printing ballots, it's enough to develop a system once. Furthermore, electronic voting is less vulnerable to corruption, administrative pressure, and human factors.

1.3 AIM AND OBJECTIVES OF THE STUDY

This project aims to develop a decentralized small-scale voting system that is secure, transparent, efficient, independently verifiable and would ultimately prevail over existing e-voting systems without decentralized structures using blockchain technology. The objectives of this study are as follows:

1. To carry out extensive research on blockchain-based voting.
2. To analyze the need for blockchain voting in electoral processes.
3. To provide a decentralized architecture to run a voting scheme that is open, fair, and independently verifiable.
4. To implement an electronic voting model that ensures voter anonymity.
5. To identify a set of current voting system gaps; potentials of the blockchain concept to improve e-voting systems.

1.4 SCOPE OF THE STUDY

This study is limited to developing an electronic voting system using blockchain. The system runs a blockchain server and collects the votes from the web user interface. Those votes are then added to the block and mined with the DPoS algorithm.

1.5 SIGNIFICANCE OF THE STUDY

This study will be of immense benefit to various electoral processes such as national elections, local elections, and elections within the university, polls, or boardroom meetings. Blockchain-based voting ensures a highly secure, decentralized, anonymous, yet auditable system that can ultimately prevent voter fraud and other electoral anomalies.

1.6 DEFINITION OF TECHNICAL TERMS

Blockchain: A system in which a record of transactions made in bitcoin or another cryptocurrency is maintained across several computers that are linked in a peer-to-peer network.

Cryptocurrency: A digital currency in which transactions are verified and records maintained by a decentralized system using cryptography, rather than a centralized authority.

Decentralization: The transfer of control of an activity or organization to several local offices or authorities rather than a single one.

Hyperledger **Fabric**: Hyperledger Fabric is a modular blockchain framework that acts as a foundation for developing blockchain-based products, solutions, and applications using plug-and-play components that are aimed for use within private enterprises.

Bitcoin: A type of digital currency in which a record of transactions is maintained and new units of currency are generated by the computational solution of mathematical problems, and which operates independently of a central bank.

Ethereum: Ethereum is a decentralized, open-source blockchain featuring smart contract functionality.

Cryptography: The art of writing or solving codes.

Node: A node is a computer that connects to a cryptocurrency network.

Blocks: Blocks are records, which together form a blockchain.

DPoS: Delegated proof of stake (DPoS) is a verification and consensus mechanism in the blockchain.

Ledger: A ledger is a database that is consensually shared and synchronized across multiple sites, institutions, or geographies, accessible by multiple people.

Mining: The process of adding transaction records to a bitcoin blockchain.

CHAPTER TWO
LITERATURE REVIEW

2.1 INTRODUCTION

To find out how blockchain technology can be incorporated in electronic voting, a literature review is carried out. Its purpose is to identify relevant research papers and use them to find and compare different solutions for blockchain-based voting.

2.1.1 OVERVIEW OF ELECTRONIC VOTING

Electronic voting (E-voting) is seen as a tool for advancing democracy, building trust in electoral management, adding credibility to election results, and increasing the overall efficiency of the electoral process (Wolf *et al.*, 2015). As technology evolves, election managers, observers, international organizations, vendors, and standards agencies are continually updating their processes and techniques. When properly implemented, e-voting solutions can eliminate certain common avenues of fraud, speed up the processing of results, increase accessibility, and make voting more convenient for citizens—in some cases, when used across a series of electoral events, potentially even lowering the long-term costs of elections or referendums. Unfortunately, not all e-voting projects succeed in delivering on such high promises. The current e-voting technology is not problem-free.

Legislative and technical challenges have risen in some cases; in others, there has been skepticism about or opposition to the introduction of new voting technologies. The inherent challenges of e-voting are considerable and linked to the complexities of electronic systems and procedures. Many e-voting solutions lack transparency for voters and even for election administrators. Most e-voting solutions are only fully understood by a small number of experts and the integrity of the electoral process relies largely on a small group of system operators instead of thousands of poll workers. If not carefully planned and designed, the introduction of e-voting can undermine confidence in the whole electoral process. It is therefore important to devote adequate time and resources to considering its introduction and looking at previous experiences of electronic voting.

The first use of the electronic voting system was in the year 2000 in the U.S.A. This was followed by France (2001), UK (2002), Spain (2003), Ireland (2004), Estonia (2005), Portugal (2005), Netherlands (2004, 2006, 2007), Paraguay (2008), Finland (2008), Austria (2009), Germany (2009) and Norway (2011). Estonia was the first country to allow remote electronic voting in the 2007 national parliamentary elections, following the use of the system during a small-scale election in 2005 (Esteve *et al.*, 2015).

2.1.2 Characteristics of Electronic Voting

According to Roh and Lee (2020), Electronic voting is implemented by applying appropriate cryptographic algorithms. The scheme must satisfy the following voting requirements:

I. Completeness: All information compiled on the ballot should be handled correctly.

II. Soundness: Votes should not be interrupted by unjust voters.

III. Privacy: No one else should know what the voter voted for.

IV. Unreusability: Voters must be able to vote only once

V. Eligibility: Voters can only vote if they have the power to vote.

VI. Fairness: No matter what happens, you should not influence your vote.

VII. Verifiability: Anyone should be able to verify the results of the voting.

2.1.3 Problems of Electronic Voting

Election manipulation is the most common concern individuals have when it comes to electronic voting. The fact is that wherever there is internet connectivity, viruses and malware are a possibility. Because the internet is such a large network, it takes a great deal of knowledge, time, and money to adequately monitor and disable all potential security risks.

Political voting would be extremely difficult because even the tiniest security flaw may be exploited, resulting in significant harm and repercussions. However, in recent years there have been an increasing number of studies that advocate using the blockchain to implement an electronic voting system to overcome these issues.

2.1.4 Blockchain in Electronic Voting

The blockchain techonolgy keeps a list of records that are very difficult to modify. Specifically, the blockchain data structure is a public distributed ledger formed by appending blocks that are linked as a chain: each block contains a hash value that is a function of the previous block (Villalobos *et al.,* 2019).

The utilisation of Blockchain technology in e-voting applications is not a new thing. Many schemes have been proposed; however, most of those protocols lack proper documentation, and questions remain about their internal workings. To date, even though blockchain is still recognized by many as an unnecessarily complicated technology, its benefits on being able to secure all the results, ensuring anonymity, and yet tampered proof, can still attract researchers to study the possibility of adopting blockchain to a voting system in practice. One of the significant studies in recent years is in (Hjlmarsson *et al.*, 2018). This study is one of the earliest which proposes the usage of a smart contract mechanism for use in an election. The preliminary architecture design is presented and discussed. The suggestion on the possibility of implementing the proposed architecture in practice is also briefly described. Another significant study is in (Yavuz *et al.*, 2015). This study addresses a more precise implementation detail on using a smart contract mechanism in designing an electronic voting system.

2.2 CONCEPTUAL FRAMEWORK

All blocks of a blockchaincomprise transactions and hash pointers, which serve as connections to the next block. It is not feasible to delete or introduce a new block in the midst of a chain in this manner since the hashes will not match. Blocks are created by network participants, who are processing transactions by running blockchain's client software. Such a participant is called a "node" (Vranken, 2017). For a block to be included in a chain all network participants must confirm its authenticity. A block is included in a

chain after a certain amount of confirmations; however, it is being confirmed by all network nodes, until everyone has an up-to-date blockchain structure. Consensus can be obtained in a variety of ways, depending on the technique employed (proof-of-work, proof-of-stake, etc.) In (Narayanan, 2016), it is mentioned that despite theoretical difficulties and high possibility of subversion, current applications of distributed consensus (Bitcoin, others) are working even better than they should, although there is not an exact, scientifically proven explanation for that. One of the main theories of blockchain is the immutability of the recorded entries. For this purpose, a wide range of other computational and mathematical tools are used, for example, Bitcoin uses hash pointers, which link the previous block to the next one, and Merkle tree roots, which allow for the efficient whole blockchain validation (Narayanan, 2016). Some blockchains employ game theory (e.g., the issue of Byzantine generals) to establish agreement inside a decentralized network where malevolent individuals have the potential to undermine the network. Exact mechanisms for achieving immutability depend on the intended application of blockchain and its parameters, such as scalability and accessibility (Croman, 2016). Another important part of blockchain technology is data distribution. Blockchain can use classical data distribution models – peer-to-peer and client-server models (Lewis, 2016), and each has consist advantages and drawbacks, which must be carefully assessed against the intended goals.Peer-to-peer data delivery, for example, will be less reliable, but it will provide more decentralization and, as a result, better protection against network closure. Distributed systems can also achieve greater computational power by combining it from users, but it comes with a cost of possible damage done by malicious users (Tama, 2017).

Some users may attempt to hack the blockchain by creating a large number of different identities to authenticate their transactions using an automated approach. There's a smart way of circumventing this difficulty, employing a notion known as proof-of-work and mining. As a result, cheating would need enormous computer resources, making it unfeasible. For the proof-of-work concept to function, network users must be rewarded for their assistance in validating transactions. The solution to this problem is to compensate those who assist in the validation of transactions.The reward of crediting them with some Bitcoin is used to encourage users on the network to assist validate transactions, despite the fact that this has now become a computationally expensive operation. Provided the Bitcoin

reward is large enough that will give them an incentive to participate in validation. In the Bitcoin protocol, this validation process is called mining (Anna-Lali, 2015).

2.3 THEORETICAL FRAMEWORK

The theoretical background gives an overview of the general concept of the project topic as seen by other researchers. Blockchain technology creates a rigid system that cannot be manipulated in order to prevent individuals from carrying out corrupt practices in electoral processes and thereby increase the trust amongst the electoral body and electorates.

Vo-Cao-Thuy *et al*., (2019) reviewed the requirements of an Ethereum based E-Voting system and then proposed "Votereum", an Electronic voting system that utilizes blockchain technology. The proposed system is empowered by the Ethereum platform which includes one server that manages the entire system and another that handles all blockchain-related requests.

Bhavani (2018) suggested the adoption of blockchain in the distribution of databases on e-voting systems can reduce one of the cheating sources of database manipulation by using the AES algorithm for encrypting data fetched from fingerprint sensors used. This research discusses the recording of voting results using blockchain algorithms from every place of election.

Khouri *et al*., (2018) proposed a novel approach for a decentralized and trustless voting platform that relies on blockchain technology to solve trust issues. The main features of this system include ensuring data integrity, transparency, and enforcing one vote per mobile phone number for every poll with ensured privacy. To accomplish this, the Ethereum Virtual Machine (EVM) is used as the Blockchain runtime environment.

Hanifatunnisa and Rahardjo (2017) developed a recording system for blockchain voting. The recording system occurs when the vote is over. The system is aimed at solving the problems that occur in the electoral system. The use of hash values in recording the voting results of each polling station linked to each other makes this recording system more secure and the use of digital signatures makes the system more reliable.

Ayed (2017) proposed a voting solution that employs a separate blockchain for each candidate. The votes to specific candidates are added to the corresponding blockchain. As the voting method is designed with official elections in mind, there is a node in each voting district.

Wu (2017) introduced a voting system that provides further anonymity to a voter by incorporating cryptographic signatures, such as ring signatures or blind signatures in the voting scheme. This particular voting system based on the Bitcoin network and ring signatures contain three main entities: a registration authority (RA), election authority (EA), and a Bitcoin address pool that consists of randomly generated Bitcoin addresses.

Koç *et al*., (2018) propose a voting system based on smart contracts on the Ethereum blockchain. Since deploying smart contracts on the official blockchain is costly, the voting process in question is carried out on Rinkeby, an Ethereum test network. To use a test network, users must obtain an official Ethereum wallet and use the settings menu to change the connection to the test network.

Tarasov (2017) proposed a voting protocol that utilizes basic functions offered by the Zcash blockchain to create a platform to cast votes. Zcash is chosen as it allows to anonymize the users who participate in an election with its option to pass both private and transparent values during a transaction. Zcash is a cryptocurrency born from the Zerocoin protocol, which was designed to obscure the trail of transactions on the Bitcoin blockchain.

Hjálmarsson (2018) evaluates the potential of distributed ledger technologies through the description of a case study and further implements a blockchain-based application that aims at improving electoral security decreasing the cost of hosting a nationwide election.

2.4 SUMMARY

Due to the growing popularity of blockchain technology, there are already multiple proposed methods and existing commercial solutions that promise secure voting with blockchain. Many researchers have explored these solutions and have agreed that this nascent technology may transform voting practices in electoral processes.

CHAPTER THREE

SYSTEM ANALYSIS AND DESIGN

3.1 INTRODUCTION

This chapter presents the system architecture, research methodology adopted and also analyzes the existing methods adopted. The research methodology includes data collection, tools used, and techniques adopted for developing the proposed system. This chapter will define the approach in achieving the development of the system while emphasizing the problem statement, and the system details such as systems architectural design, user interfaces and data management.

3.2 RESEARCH METHODOLOGY

The methodology offers an overview of the research design, types of data to be collected, sampling design, and relevant interpretation towards the conducting of respective research, and necessary tools selected for proposed hypotheses of the research. This thesis aims at exploring blockchain technology in voting systems.

3.2.1 Data Collection Methods

During the research work, data collection was limited to the internet. Due to the unavailability of primary data, secondary data was used. A thorough search of published materials was conducted, followed by the selection of relevant sources based on criteria. Finally, the studies were examined and categorized, with the extracted data synthesized and summarized.

3.3 SYSTEM ANALYSIS

System analysis is a problem solving technique that involves understanding the process involved, identifying problems and recommending feasible suggestions for improving the system functioning. This involves studying the business process, gathering operational data,

understanding the information flow, finding out bottlenecks and evolving solutions for overcoming the weaknesses of the system so as to achieve the organizational goals.

3.3.1 Analysis of the Case Study

The voting mechanism in Nigeria and a host of other nations is based on paper ballots. During the voting phase, the voter selects her option(s) on a ballot form in a booth, then departs and casts the marked ballot form in a ballot box. An official guarantees that she casts just one ballot form, but she should not be able to discover the voter's vote choice or link the voter to the ballot form she cast. The voting booth, ballot form, and ballot box are the three essential components here. After the polls have closed, the tallying process begins. Ballot boxes are gathered and taken to a specified tallying location. A crew of observers watches officials open the ballot boxes and count the ballots. The final tally is released when the local results are compiled.

Even though its weakness has been exploited throughout the years, this voting system is still in use in the majority of African countries today. Due to the high expense of development and implementation, several countries find it impossible to adopt any new voting system and instead focus on improving the issues that are vulnerable to the ballot-box method.

3.3.2 Weaknesses of the Present System

The traditional voting system has been found to have the following weaknesses that are enumerated below:

1. Integrity: The whole system is based on the trustworthiness of the persons in charge of it. This makes people more vulnerable to corruption since they can be bribed.
2. Accessibility: Polling station sites can be a barrier for voters who reside in remote regions, as well as voters who may be unable to physically attend a polling station due to disability or citizens who are out of the country
3. Cost: The costs of a paper ballot method include logistic costs such as physically producing the paper used, transportation, security, voting locations, and staff work.

4. Inefficiency: On a Large scale, managing a traditional paper ballot system requires an enormous amount of time and effort. Also, because paper ballots are fault tolerant, a large number of them will not be counted.
5. Violent clashes in polling stations
6. Ballot-box stuffing

3.3.3 System Justification

Blockchain proves to be the protocol that fits the bill for voting security and integrity. Votes are recorded on a shared ledger and audit trail in this system, with data that is cryptographically secure and immutable. Once a vote is recorded, it cannot be changed, and the entire voting history is easily traceable. Voters could have complete trust in the shared data because it would be transparent and confidential, i.e. sensitive voter registration data would be kept off-chain and linked to ledger blocks using cryptographic hashes.

3.4 ANALYSIS OF THE PROPOSED SYSTEM

Considering the limitation and challenges with the existing system, there is a need to design an electronic voting system incorporated with blockchain. The system is very flexible and easy to use. It has features that ensure voting/conduct of elections is fast and productive.

The proposed system designed in this paper is web based and is of the one voter, one vote principle where each voter is only eligible to cast just a single vote in the voting process. As seen in figure 1 below, votes are registered in transactions where the voter transfers tokens into the candidate's account.



Figure 1: Blockchain Interfaced Sacured E-Voting System.
Source: Shrestha *et al.* (2018)

3.4.1 Input Analysis

After conducting extensive research to discover facts about the functionality of the blockchain voting system, these data are analyzed or evaluated, and the outcome is an informal description of the system under study. This informal description will form the basis for the design of the new system. As a result, it is vital that the acquired data be properly analyzed.

The input analysis deals with the process used to feed data to the system for processing. Here, data is manually fed in with a keyboard and/or a mouse. The data supplied to the system includes the following:

I. Account ID
II. Vote choice

3.4.2 Process Analysis

This process entails combining different modules and code to obtain the desired output. Some of the modules include the login module which contains a link to the voting booth that allows the user access to vote.

3.4.3 Output Analysis

Computer system outputs are generally used to communicate the process's outcome to users. They are also utilized to keep a permanent duplicate of the result for future reference. For users, the output is the most essential and immediate source of information.

It is envisaged that many positive changes in the voting process will occur as a result of the implementation of this new system. In the design of the electronic-based voting system, conscious effort is made to create an effective web bases system that would be successfully adopted in electoral processes.

3.4.4 Benefit of the Proposed System

The proposed system has been found to have a number of advantages over the existing system, some of the advantages are:

1. Transparency: By using blockchain, Votes can be counted and kept on an immutable public ledger. This means they may be numbered and tracked while remaining visible to all. As a result, by allowing voters to see real-time records of the amount of votes cast, everyone will be able to see the voting's legality, resulting in a transparent and reliable voting system.
2. Anonimity: When voting, the blockchain allows for anonymity. Voters can utilize their private keys to remain anonymous, just like transactions on the blockchain. They can then vote anonymously in the system without fear of others finding out what they voted for. The capacity to ensure anonymity may encourage more people to participate in and use the voting mechanism.
3. Security: Security is one of the most crucial aspects of voting. At the moment, voting systems are extremely vulnerable to hacking. Malicious actors can enter the system and change the outcome if there are no enough security measures in place. This is where the blockchain technology comes into play, The technology has the potential to create a system that appears to be impenetrable.
4. Process Time: It takes a long time for current voting methods to collect and process responses. It might be difficult to obtain all of the information quickly and properly when voting stations are spread out and offices are not all in one location. However, blockchain has the potential to change everything. Rather than waiting for a huge number of people to interact manually, all organizers will be able to observe the conclusion on the blockchain immediately. After the vote is completed, the results can be immediately gathered and processed.
5. Individual Verifiability: Because of the public nature of the ledger, each voter can verify that his or her vote has been entered into the blockchain and so counted. Each voter is also in charge of counting the votes, ensuring that their vote is included in the final tally.

6. Universal Verifiability: Because the ledger is open to the public, any voter can check that the votes were counted correctly by counting them. External auditors can also verify the results by obtaining a copy of the blockchain and ensuring that the votes contained within it are legitimate.

3.5 SYSTEM DESIGN

The proposed system was designed using various web technologies like Javascript, REACT framework and node. The development of this system is divided into different phases which are the frontend and backend.

The frontend consists of the following technologies:

- HTML
- CSS
- JAVASCRIPT
- REACT

The Backend part of the program was developed using:

- NODE JS
- TYPESCRIPT

3.5.1 Architecture Diagram of the Proposed System

Architectural design entails understanding how a system should be organized and the appropriate development of the system's general structure. It serves as a vital link between design and requirements engineering by identifying the most important structural components of a system and their relationships, as well as providing an overview of the new system, including the most crucial parts of the system, the services they provide, and their communication.

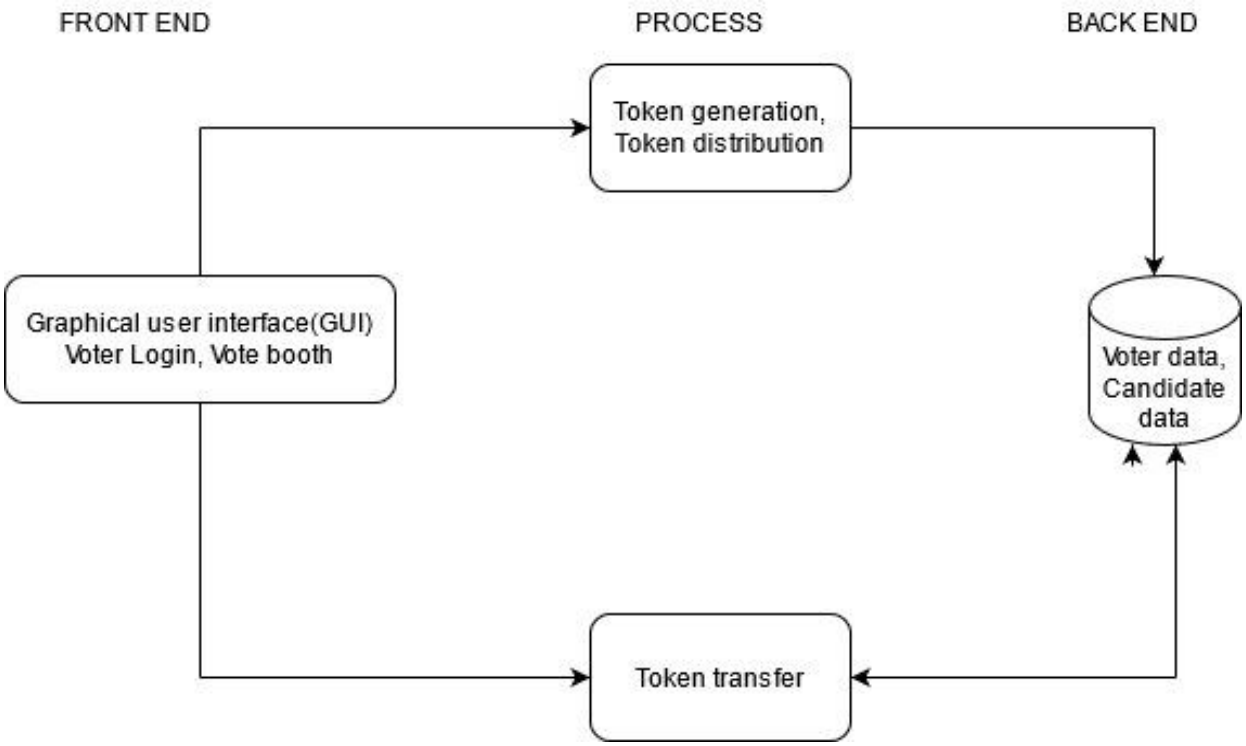


Figure 2: System Architecture.

3.6 FUNCTIONALITY OF THE SYSTEM (USE CASE DIAGRAM)

The interactions between the system and the user are graphically depicted in a use case diagram. Use case diagrams are important in system design because they serve as a roadmap for designing the system's structure, as well as defining who will use the system and how the user expects to interact with it. The Use Case diagram for the system is shown in the figure below.

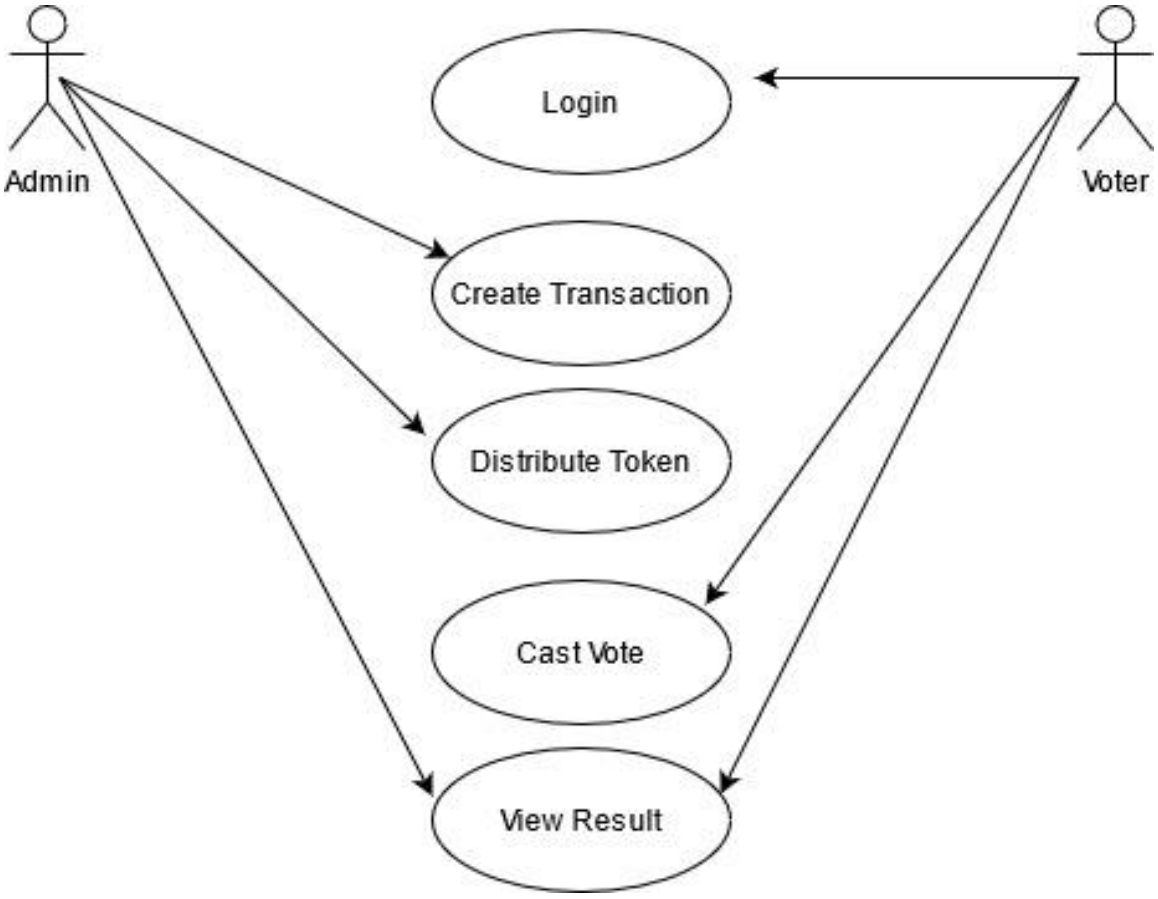


Figure 3: Use Case Diagram.

CHAPTER FOUR

PROGRAMMING, IMPLEMENTATION AND DOCUMENTATION

4.1 INTRODUCTION TO PROGRAMMING CONCEPT

Programming is the process of designing and building an executable computer program to accomplish a specific computing result. Programming involves tasks such as: analysis, generating algorithms' accuracy and resource consumption, and the implementation of algorithms in a chosen programming language (Hartmann, 2013). Languages range from low level to high level. Lower level languages are more intimately related to the platform they're designed for, whereas higher level languages abstract the platform from the programmer to a greater extent.

Web application programming or development concepts are implemented to develop a web-based blockchain voting system in this study. Web application programming is the development of application programs that are hosted on remote servers and delivered to a user's device via the Internet. A web application does not require downloading, it is instead accessed over the internet via web browsers such as Google Chrome, Safari, or Mozilla Firefox. A majority of web applications can be written in HTML5, Cascading Style Sheet (CSS) and JavaScript.

Web application programming is typically divided into two phases namely; Front-end development and Back-end development. Front-end development for web applications is accomplished through client-side programming.Client-side programming utilizes HTML, CSS and JavaScript. HTML is the standard markup language for creating web pages which describes the structure of the web page, CSS describes the presentation of web pages, including colors, layout, and fonts while JavaScript makes the content of web pages interactive.

The Server-side programming which is the back-end powers the client-side and is used to create the scripts that web applications use. Scripts could be written in JavaScript,

Typescript, Python and a host of other programming languages. Server-side scripting creates a custom interface for the end-user and hides the source code that makes up the interface.

4.1.1 Software Methodology

This refers to the steps and procedures that must be followed in order to realize the desired model. Each of the following processes is distinct and crucial for the development of the application, as each state is dependent on the others in some way. The software methodology used for this program is the waterfall model and below are the stages of implementing the development model.

I. Identifying and Understanding the problem
II. Program design
III. Coding
IV. Debugging
V. Testing
VI. Documentation

4.2 Identifying and Understanding the problem

The longstanding issues of voter fraud, inefficiency, ballot-box snatching and many other anomalies in electoral practices have proven to be major issues in elections. Also, most elections require voters to physically report to a polling booth and cast their ballot with a paper vote. These led to the development of this system that is remote, secure, decentralized, and transparent and also processes results immediately after the election is completed.

4.2.1 Program Design

This i**s** the next stage done to achieve the program. Program design is a process to transform user requirements into some suitable form, which helps the programmer in software coding and implementation. In this program, two screens are used in the design, they are:

I. Input Screen Design: This interface will be used for casting the vote, it contains the details of the electoral candidates and it features a login page where voters would be able to login to their accounts.

II. Output Screen Design: This is designed to show the display screen after the user successfully casts a vote.

4.2.2 Coding

Blockchain technology was incorporated in building the web based voting system. Typescript and some JavaScript frameworks such as React and Node were used in the development of the web application. HTML, CSS and JavaScript are implemented in the frontend part of the application while TypeScript is implemented in the backend.

4.2.3 Debugging

Some bugs were encountered in the program in cases such as trying to integrate the blockchain contract into the application and also when I tried to run the program after successful integration. To counter the bugs, I installed the necessary node package managers on my system and the program ran successfully.

4.2.4 Testing

Unit and integration testing was carried out for the program designed, the program was tested by first loading all the modules in the program one by one so as to prevent errors and determine if they are fit for use and subsequently loading all the modules together to check for program efficiency.

4.2.5 Documentation

This is usually the last stage of application development and it is an essential step in the program development, though most programmers neglect this stage by giving many reasons. It provides users and the programmer the necessary information to maintain the software in future, and also may help the programmer to correct problems that may occur in the program.

4.3 DEVELOPMENT LANGUAGES/TOOLS USED

The study used JavaScript as the main programming language in the development of the blockchain voting application. Below is a list of languages and libraries/frameworks that were used in achieving the research objectives

JAVASCRIPT: JavaScript is a high-level lightweight, interpreted, event driven, and multi-paradigm programming language. It is designed for creating network-centric applications. It is very easy to implement because it is integrated with HTML. It is open-source and cross-platform.

NODE JS: Node.js is an open-source, cross-platform, back-end JavaScript runtime environment that runs on the V8 engine and executes JavaScript code outside a web browser.

REACT JS: React is a declarative, efficient, and flexible open-source front-end JavaScript library for building user interfaces or UI components.

4.4 JUSTIFICATION OF PROGRAMMING LANGUAGE USED

JavaScript is a high-level lightweight, interpreted, event driven, and multi-paradigm programming language. The importance of JavaScript as a web technology can be determined from the fact that it is currently used by 94.5% of all websites. As a client-side programming language, JavaScript helps web developers to make web pages dynamic and interactive by implementing custom client-side scripts. At the same time, developers can also use cross-platform runtime engines like Node.js to write server-side code in JavaScript.

4.5 TESTING

Software testing is a process of determining whether the actual software product meets the expected requirements and ensuring that the software product is free of defects. It entails running software/system components through their paces using manual or automated techniques to evaluate one or more properties of interest. The goal of software testing is to find mistakes, gaps, or missing requirements in comparison to the actual requirements. Software testing is important because if there are any defects or problems in the software, they may be found early and fixed before the software product is delivered. A well tested

software product assures dependability, security, and excellent performance, which leads to time savings, cost effectiveness, and customer pleasure.

Unit and integration testing were carried out for the program designed, the program was tested by first loading all the modules in the program one by one so as to prevent errors and determine if they are fit for use and subsequently loading all the modules together to check for program efficiency.

4.7 PRINT SCREEN OF INTERFACES

1. Login page:

This is the interface where the voters get to login with their private keys to be able to cast votes. The figure below shows the login page.

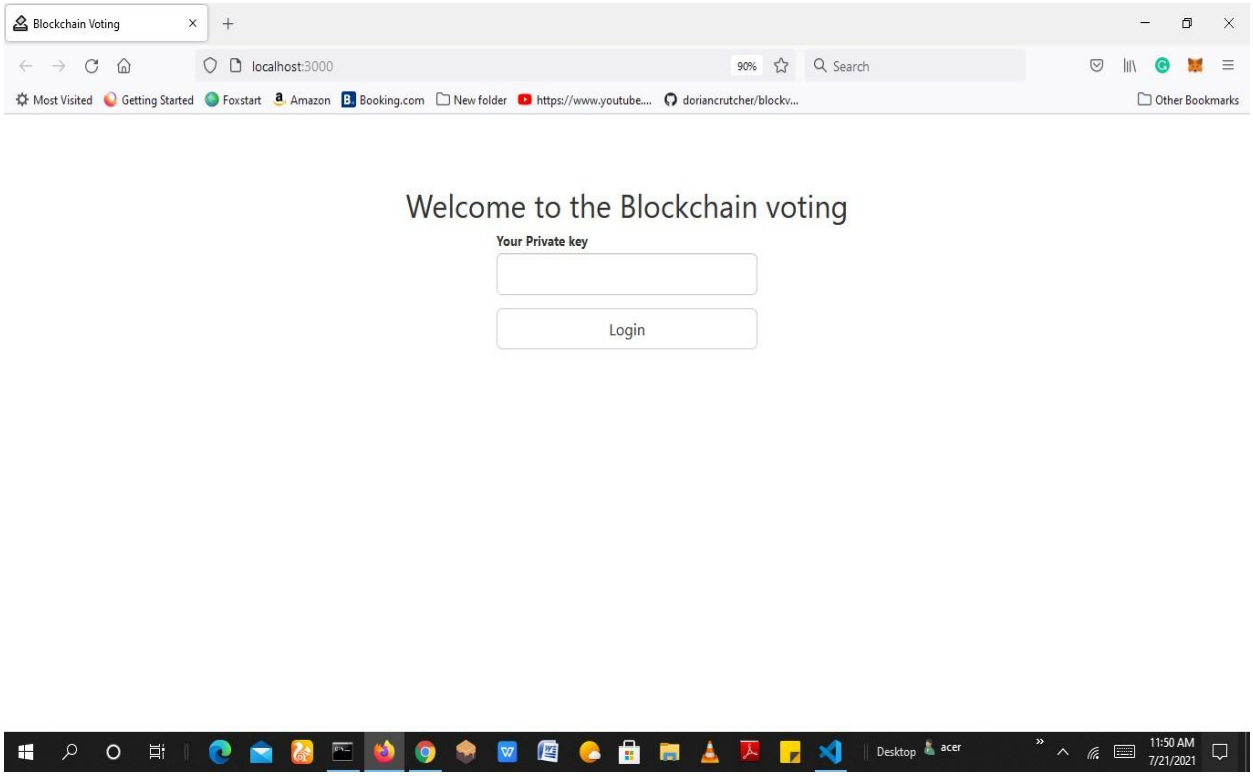


Figure 4: Login Page.

2. The Main page/Vote booth:

This is the interface where votes will be casted. It contains the respective information(s) of the electoral candidates. The vote booth is shown in the figure below:

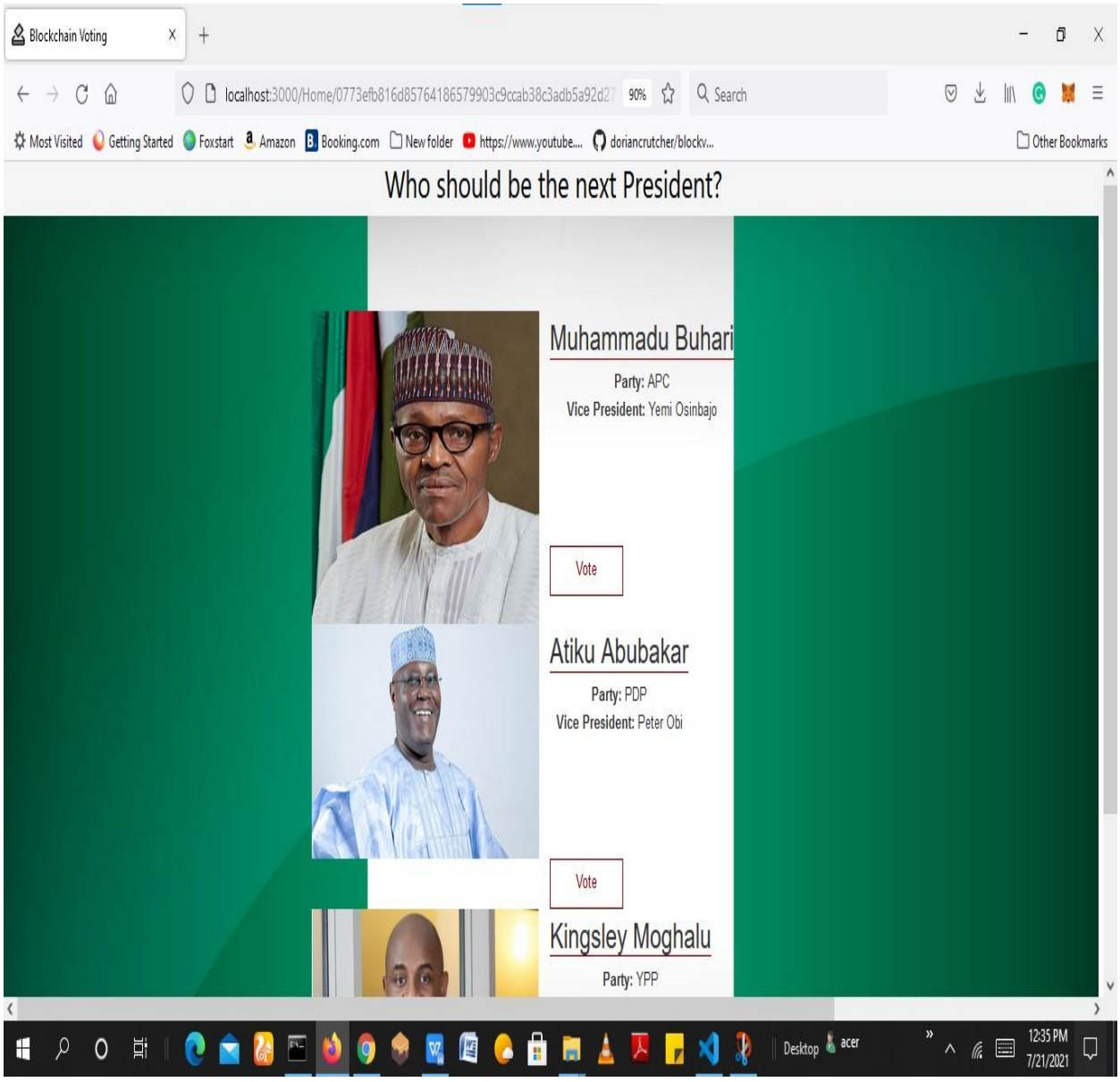


Figure 5: Vote booth.

Process Page:

This page shows the process of token distribution across nodes on the blockchain.

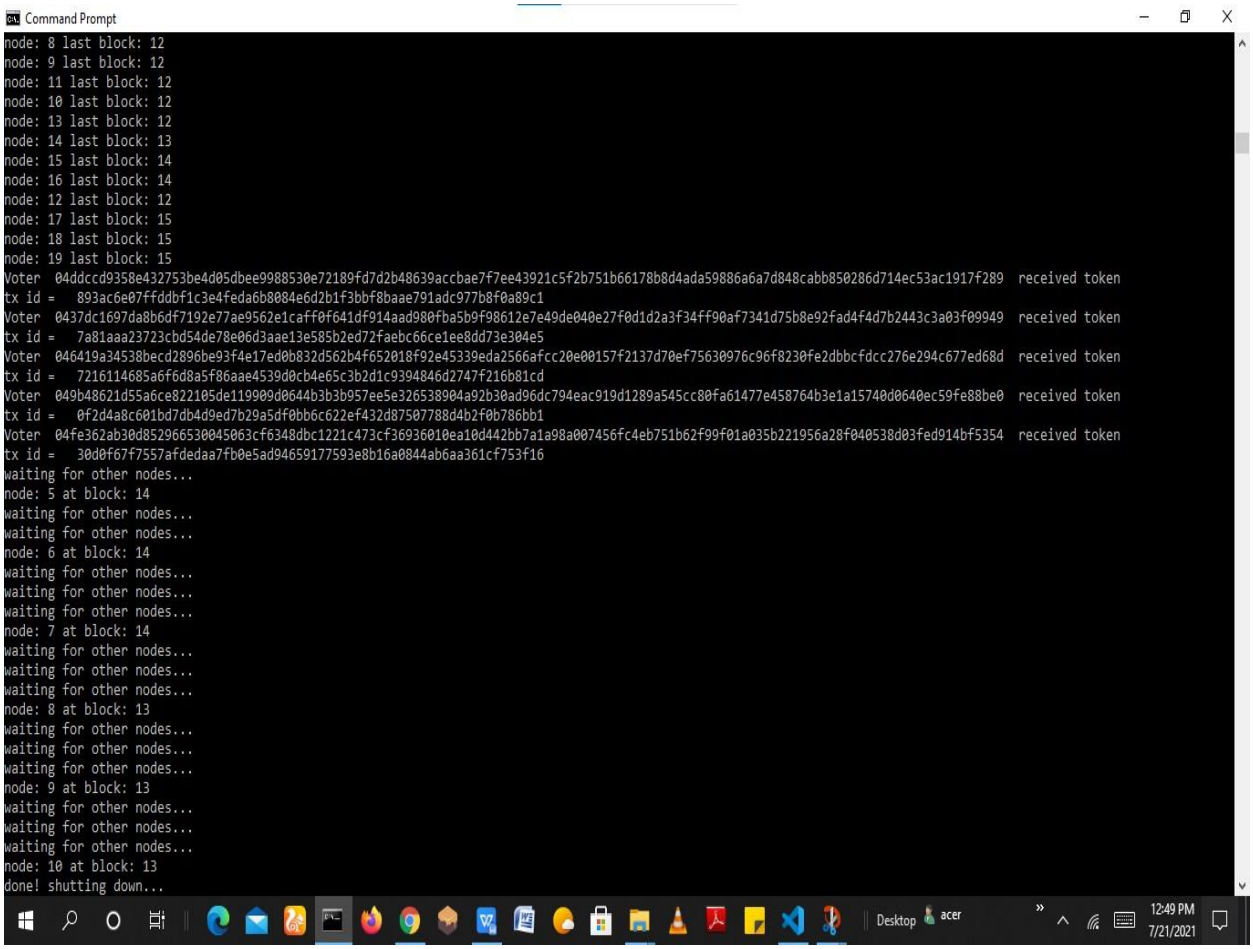


Figure 6: Token distribution.

3. Output Page:

This page shows the screen display after the voter successfully casts his/her vote. Figure 7 below shows the output age.

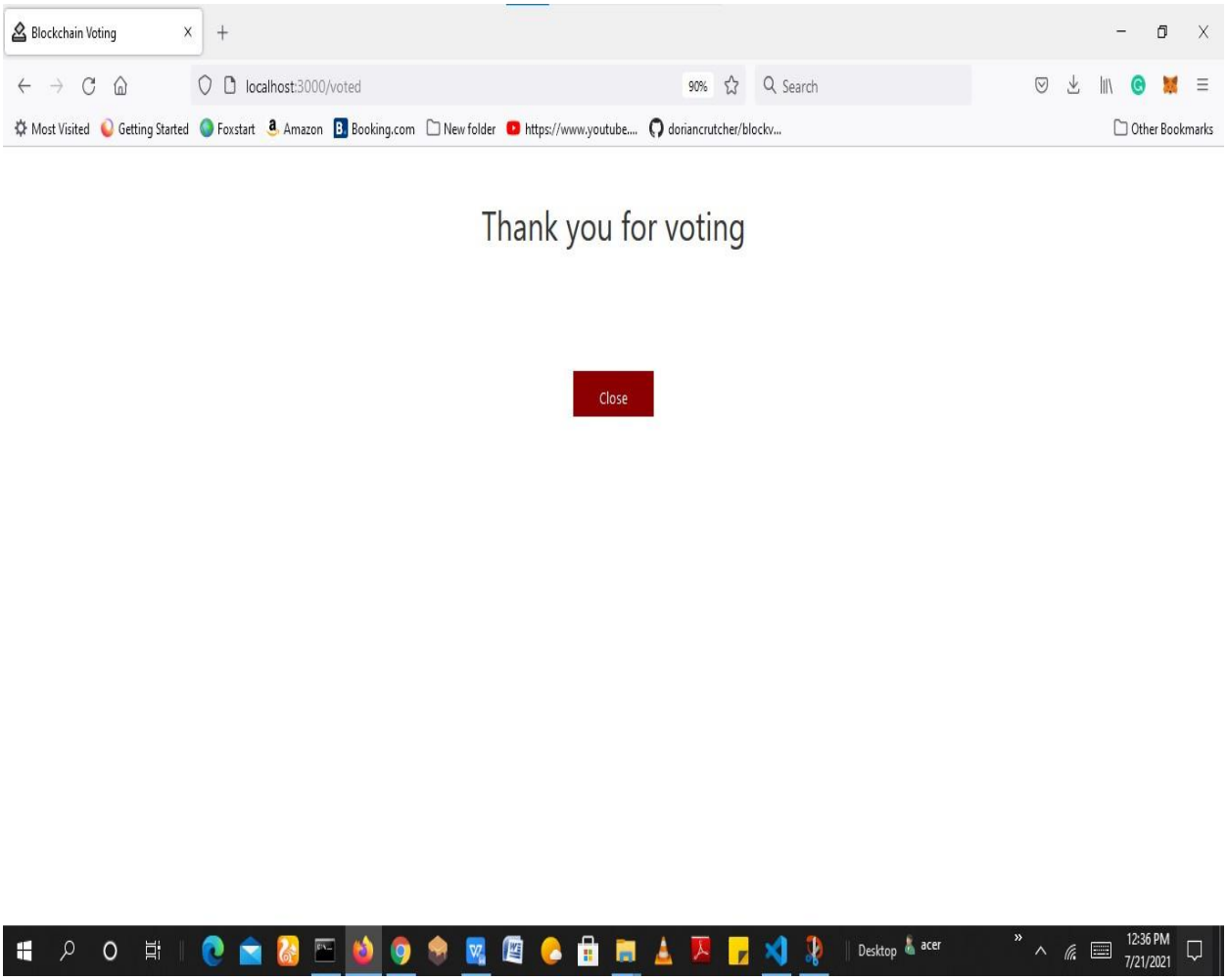


Figure 7: Output Page.

4.8 SYSTEM REQUIREMENTS

The system requirements provide an overview of the least criteria the hardware and software must obtain before its usage. It is to provide a detailed overview of the hardware and software product, its parameters and goals to the wide range of users. This describes the project's target audience and it's user interface, hardware and software requirements.

4.8.1 Hardware Requirement

The hardware requirements are as follows:

1. A minimum of 650GB hard disk drive.
2. At least a Core i5 1.6 GHz Intel Processor.
3. Minimum of 4GB Random Access Memory.
4. A CD-ROM Drive.
5. A super Video Graphic Adapter (SVGA) Monitor.
6. A keyboard and mouse.

4.8.2 Software Requirements

The minimum software requirements for running this application are:

1. Microsoft Windows (8,8.1 and 10)
2. JavaScript ES6 and above
3. Web Browsers
4. NODE JS
5. Command Prompt.

4.8.3 Running and Accessing the Program

Since it is a web application, the program could be accessed by either:

1. Web hosting: Buying a unique domain name and uploading all the file contents into a web server provided by an internet service provider.
2. Running on localhost: This could be achieved with the command prompt terminal. All blockchain scripts should be run from the root directory, and the front end scripts should be run from the subdirectory(front-end) by running the command “npm start”

4.9 Change Over Strategy

The program should be implemented using a parallel changeover which means running the new system concurrently with the old system for a prearranged period of time thereby slowly integrating the new system and getting rid of the previous system.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 SUMMARY

Blockchain is an emerging technology with the ability to track and trace transaction or product path from source to sink with exceptional security and transparency. Though, Blockchain was initially used in cryptocurrency transaction such as bitcoin, however with the rise of blockchain based applications, the potential benefit of this ground breaking technology is unparalleled. It can be observed from this research that blockchain can be applied to improve electoral processes in Nigeria, increasing efficiency and transparency. However, blockchain is a disruptive technology, hence, it is essential that effective planning and research is carried out prior to applying blockchain to government verticals. Governments of emerging economies also need to stimulate entrepreneurial awareness of emerging technology through training and promotion to facilitate investment readiness. Similar to every emerging technology, the adoption of blockchain to its full potential is dependent on the extent that government and relevant stakeholders take the lead in supporting and unveiling market-creating innovation on blockchain platforms.

5.2 CONCLUSION

Blockchain is an innovative technology capable of disrupting existing business models in a profound manner. Some of the basic characteristics of blockchain's building blocks and smart contracts which are programs that assist in the carrying out of transactions on blockchain were explained in this study. The significant benefits of blockchain generally present results in market and business factors demonstrating considerable interest in grasping and integrating this technology within their operations. The use of blockchain in voting is extremely important for the purpose of regaining the trust of citizens and the constantly evolving expectations of the general public.

5.3 RECOMMENDATION

Governments and electoral commissions should look to the successes of blockchain in various industries, learn from the experiences gained and carry out a cost/benefit analysis of upgrading voting systems nationwide using blockchain.

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